

Physics I: Comprehensive Midterm Notes

Topic 1: Electrostatics Basics (Week 1)

Key Concepts

- **Electric Charge:** Fundamental property of matter. Understand nature, quantization, and conservation.
- **Vector Superposition Principle:** Essential for solving problems with 3 or more charges. To find net force, add individual force vectors.

High-Yield Formulas

- **Coulomb's Law:** $F = k * (|q_1 * q_2| / r^2)$
 - Coulomb's constant $k = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$
 - Permittivity of free space $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 / (\text{N} \cdot \text{m}^2)$
 - Relationship: $k = 1 / (4\pi\epsilon_0)$
- **Quantization of Charge:** $q = n * e$
 - Elementary charge $e = 1.602 \times 10^{-19} \text{ C}$

Exam Practice Focus

- Calculate the force between two point charges separated by a distance.
- Find the net force on a charge at the center of a triangle/square of charges.
- Calculate how many electrons are in a given amount of Coulomb charge.

Pro Tip: Force is a **VECTOR**. If charges are in 2D, you **must** resolve them into x and y components before adding.

Topic 2: Fields, Potential & Dipoles (Week 2)

Key Concepts

- **Electric Fields:** The space surrounding charges visualized by field lines. It represents force per unit charge.
- **Electric Potential vs Potential Energy:** Potential is a scalar representation of the electric field's effect (energy per unit charge).
- **Electric Dipoles:** A pair of equal and opposite charges.
- **Dielectrics:** Insulators that can be polarized by an applied electric field, reducing the internal field.

High-Yield Formulas

- **Electric Field (Point Charge):** $E = k * (|q| / r^2)$ (Vector)
- **Electric Potential:** $V = k * (q / r)$ (Scalar)
- **Field-Potential Relation:** $E = -\nabla V = -(dV/dx \ i + dV/dy \ j + dV/dz \ k)$
- **Dipole Moment:** $p = q * d$ (Vector pointing from -q to +q)
- **Dipole Potential:** $V = (k * p * \cos\theta) / r^2$

Exam Practice Focus

- Find the electric field at a point between two opposite charges.
- Calculate the potential at a specific distance from a dipole.
- Explain how a dielectric material changes the electric field inside a capacitor.

***Pro Tip:** Potential is a **SCALAR**. You can just add the values together without worrying about angles! Remember to include the **sign** (+ or -) of the charge when calculating scalar potential.*

Topic 3: Gauss's Law (Week 3)

Key Concepts

- **Electric Flux (Φ_E):** Measure of the electric field passing through a given surface. $\Phi_E = E * A * \cos\theta$
- **Gaussian Surface:** A closed imaginary surface in 3D space through which the flux is calculated.

High-Yield Formulas

- **Gauss's Law:** $\Phi_E = \oint E \cdot dA = Q_{enc} / \epsilon_0$

The "Big 4" Derivations to Memorize

*You **MUST** be able to derive E for these from scratch using Gauss's Law:*

- **Point Charge:** $E = kQ / r^2$ (Spherical surface)
- **Infinite Line Charge:** $E = \lambda / (2\pi\epsilon_0 r)$ (Cylindrical surface)
- **Infinite Sheet of Charge:** $E = \sigma / (2\epsilon_0)$ (Pillbox surface)
- **Uniformly Charged Sphere:**

- Outside ($r \geq R$): $E = kQ / r^2$
- Inside ($r < R$): $E = (kQ / R^3) * r$

Exam Practice Focus

- Define Electric Flux and its units ($N \cdot m^2 / C$).
- Derive the electric field for an infinite sheet of charge using a "pillbox" Gaussian surface.
- Calculate the flux through a cube with a charge at its center.

Pro Tip: Choosing the right **Gaussian Surface** is the key. For lines, use a cylinder. For sheets, use a pillbox. For spheres, use a sphere. Always exploit symmetry!

Topic 4: Intro to Networks (Week 4)

Key Concepts

- **Networks:** Master the "Divider Rules" to save time during circuit calculations.
- **Series & Parallel Resistors:** How to simplify complex networks into a single equivalent resistance.

High-Yield Formulas

- **Ohm's Law:** $V = I * R$
- **Equivalent Resistance:**
 - Series: $R_{eq} = R1 + R2 + \dots + Rn$
 - Parallel: $1/R_{eq} = 1/R1 + 1/R2 + \dots + 1/Rn$ (For two: $R_{eq} = (R1 * R2) / (R1 + R2)$)
- **Voltage Divider Rule (VDR - Series):** $V_x = V_s * (R_x / R_{eq})$
- **Current Divider Rule (CDR - Two Parallel Resistors):** $I_1 = I_s * (R_2 / (R_1 + R_2))$ *(Note: The OTHER resistor is on top!)*
- **Power:** $P = V * I = I^2 R = V^2 / R$

Exam Practice Focus

- Simplify a complex network of 5-6 resistors into one R_{eq} .
- Find the voltage across a specific resistor using VDR.
- Find the current through a parallel branch using CDR.

Pro Tip: In CDR, the current through one branch is proportional to the **other** branch's resistance. Don't mix them up!



Topic 5: Advanced Circuit Analysis

(Week 5)

Key Concepts

- **Kirchhoff's Voltage Law (KVL - Loop Rule):** Sum of voltage drops and rises in a closed loop equals zero ($\sum V = 0$). Based on Conservation of Energy.
- **Kirchhoff's Current Law (KCL - Junction Rule):** Sum of currents entering a junction equals sum of currents leaving ($\sum I_{in} = \sum I_{out}$). Based on Conservation of Charge.
- **Mesh Analysis:** Solving circuits using loop currents and KVL.
- **Nodal Analysis:** Solving circuits using node voltages and KCL.

Key Analytical Steps

- **Mesh Analysis:**
 - Assign loop currents ($i_1, i_2...$) in the same direction (e.g., clockwise).
 - Write KVL equations for each loop.
 - Solve the system of linear equations.
- **Nodal Analysis:**
 - Pick a Ground (0V) reference node.
 - Assign node voltages ($V_1, V_2...$) to the other junctions.
 - Write KCL equations at each non-reference node.

Exam Practice Focus

- Solve a 2-loop circuit using Mesh Analysis.
- Find the node voltages in a circuit with 3 nodes.
- Apply Kirchhoff's Laws to find an unknown current in a bridge circuit.

Pro Tip: If a circuit has many parallel branches, **Nodal Analysis** is usually faster. If it has many series loops, **Mesh Analysis** is better. Picking the right one saves you 10 minutes in the exam!

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